

Dušan Korizma

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EDUCATION

Mathematical Grammar School

Graduated with honours

Belgrade, Serbia

Sept. 2019 – Jun. 2023

University of Belgrade, School of Electrical Engineering

Software Engineering, Current GPA: 9.63

Belgrade, Serbia

Oct. 2023 – Sept. 2027

EXPERIENCE

Petnica Science Center

Junior Teaching Assistant

Oct. 2023 – Present

Valjevo, Serbia

- Creating and organizing lectures and workshops
- Mentoring Computer Science projects for high school students

University of Belgrade, School of Electrical Engineering

Junior Teaching Assistant

Nov. 2025 – Present

Belgrade, Serbia

- Assisted teaching staff in organizing laboratory sessions, grading assignments, and evaluating students
- Supported instruction across 5+ courses, including Object-Oriented Programming, Databases, and Data Structures and Algorithms

PROJECTS

Simple RISC-V Operating System

- Implemented the kernel for a simple operating system
- Functionalities included asynchronous context switching, semaphores, user threads and memory management
- Implemented daemon threads for asynchronous input/output buffer management.

Analysis of Genetic Algorithm Method for Deriving a Function Formula from a Given Graph

- Developed a Python-based system to extrapolate mathematical formulas from graphical data using binary trees and the genetic algorithm
- Combined evolutionary optimization and tree-based representations to accurately derive relationships from complex datasets

Memory Allocator Upgrade for the xv6 Operating System

- Implemented a thread-safe physical memory allocator for xv6 as an alternative to the default allocator
- Built a buddy allocator to efficiently manage variable-sized page allocations and reduce fragmentation
- Added a slab allocator for frequent kernel object allocation, improving allocation efficiency for common structures

Command Line Interpreter

- Implemented a Command Line Interpreter for a subset of UNIX commands
- Implemented input/output redirection and piping
- Emphasized detailed planning and thoughtful class relationship design, highlighting principles of object-oriented architecture

SQLike Interpreter

- Built an end-to-end SQL-like interpreter with lexer, parser, AST construction, intermediate representation, and query execution
- Modeled the internal query representation after PostgreSQL's planner/interpreter architecture
- Designed an extensible execution layer to support future query optimization techniques

SKILLS

Skills: C, C++, Python, Git, SQL, Java, x86, Go, Docker